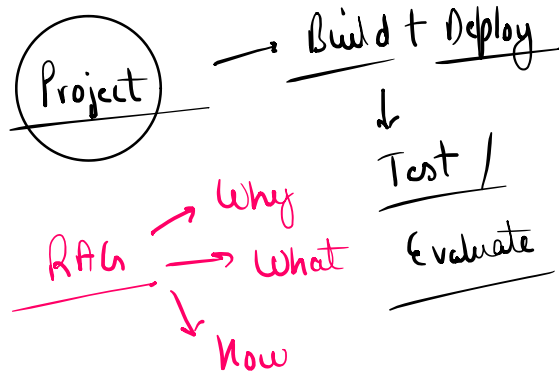




Components :-

- ① Document loaders.
- ② Text splitters.
- ③ Embedding model
- ④ Vector stores
- ⑤ Retrievers.
- ⑥ Advanced Retrievers.

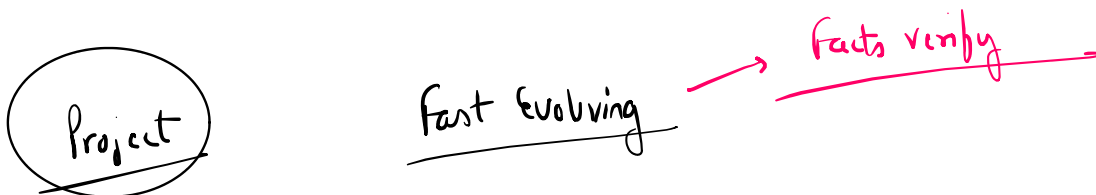
Advanced RAG pipelines

CRAI, Self-RAG,

HyDE, Agentic RAG,

Graph RAG → Knowledge graphs

RAG Evaluation : → RAGAs → 5 diff metrics



Gen AI Research

① Knowledge

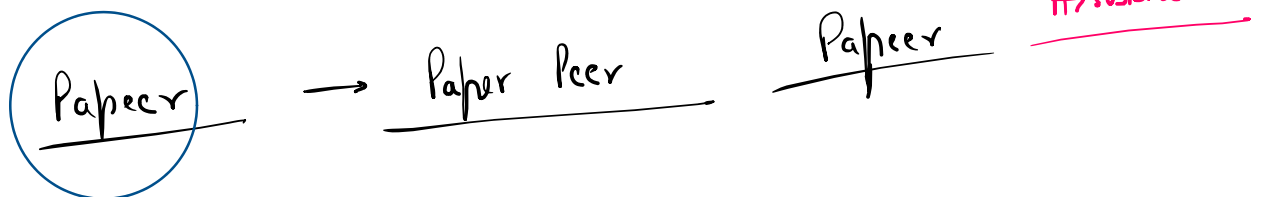
② Claims / Facts verify

Fast Evolving

Facts verify

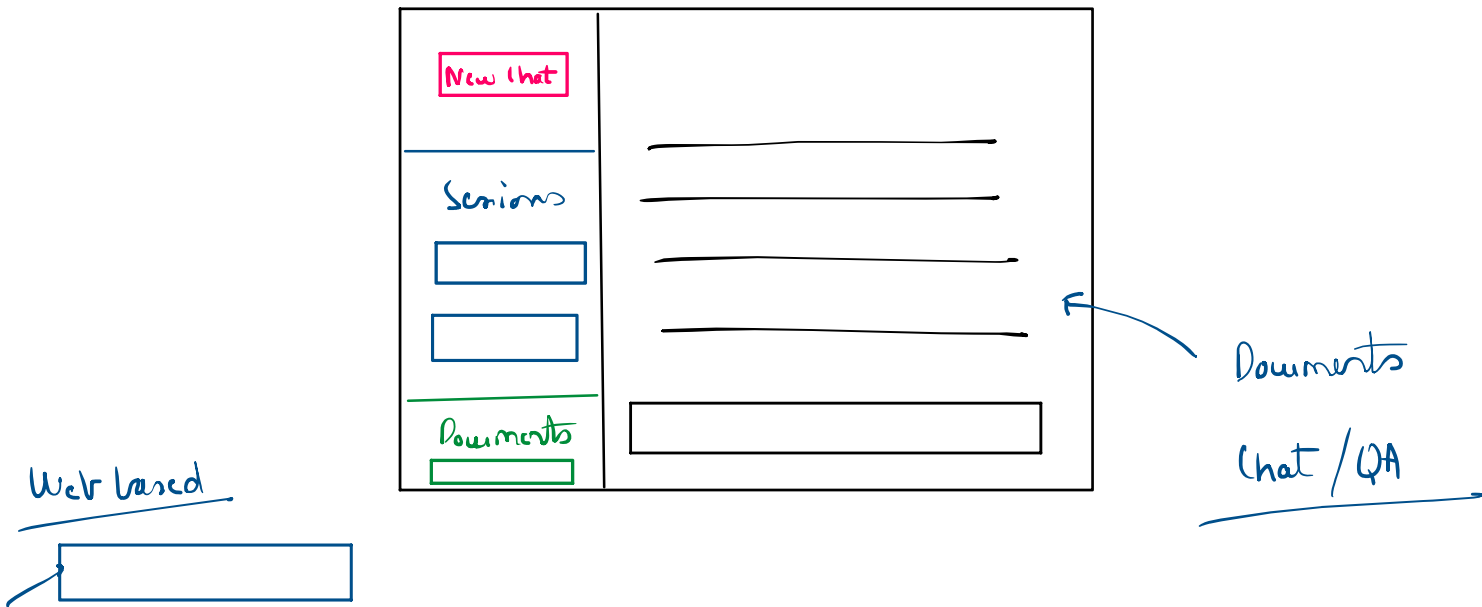
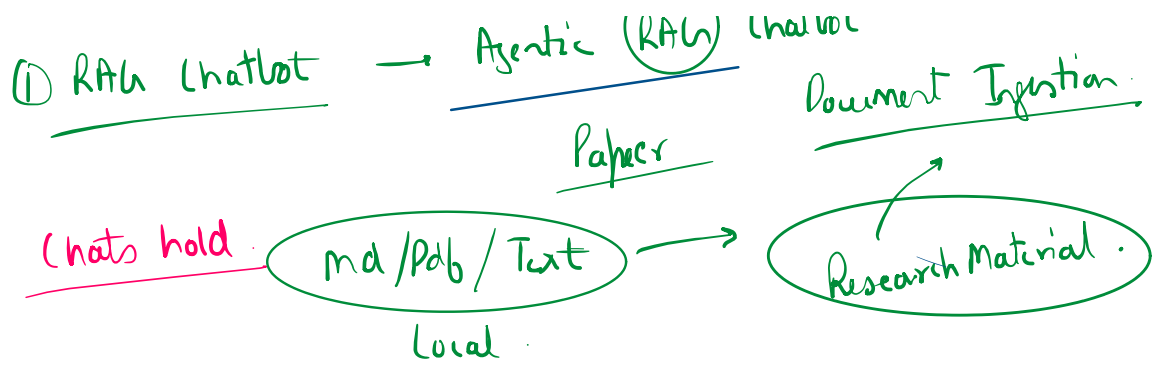
↓
LIMS → Transformer arch.

outdated



① RAG Chatbot

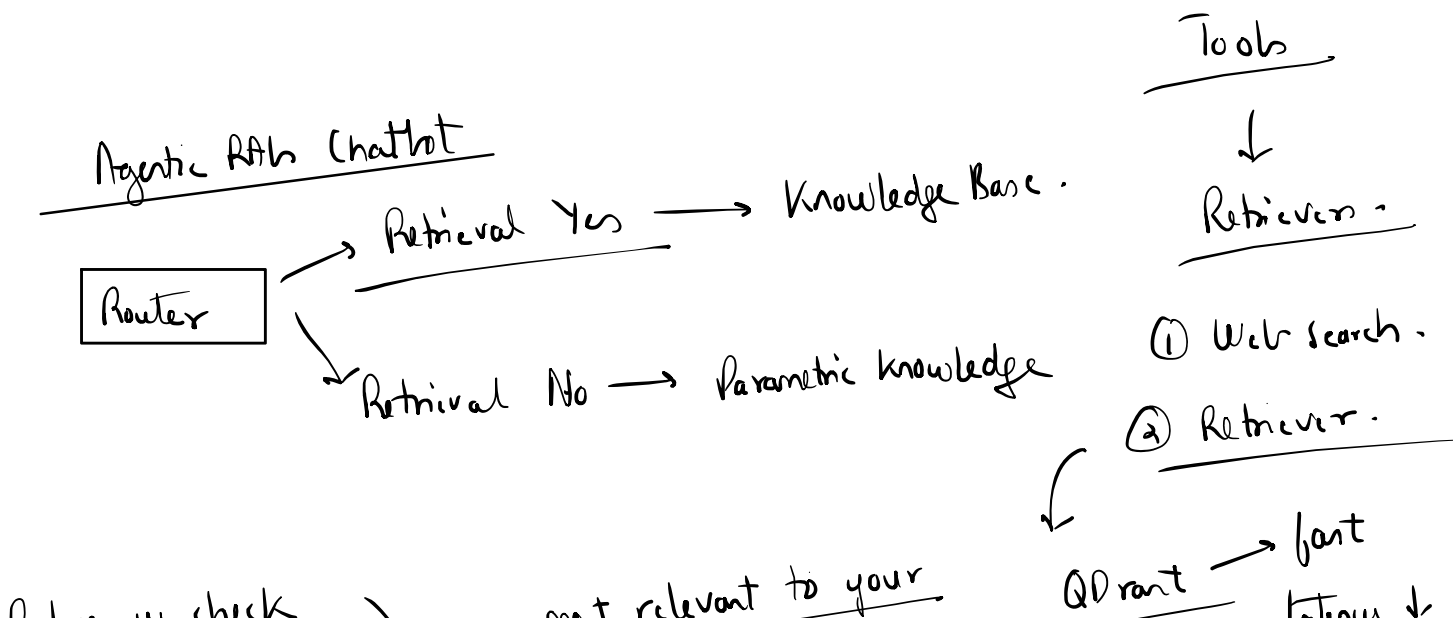
→ Agentic RAG Chatbot
Document Ingestion

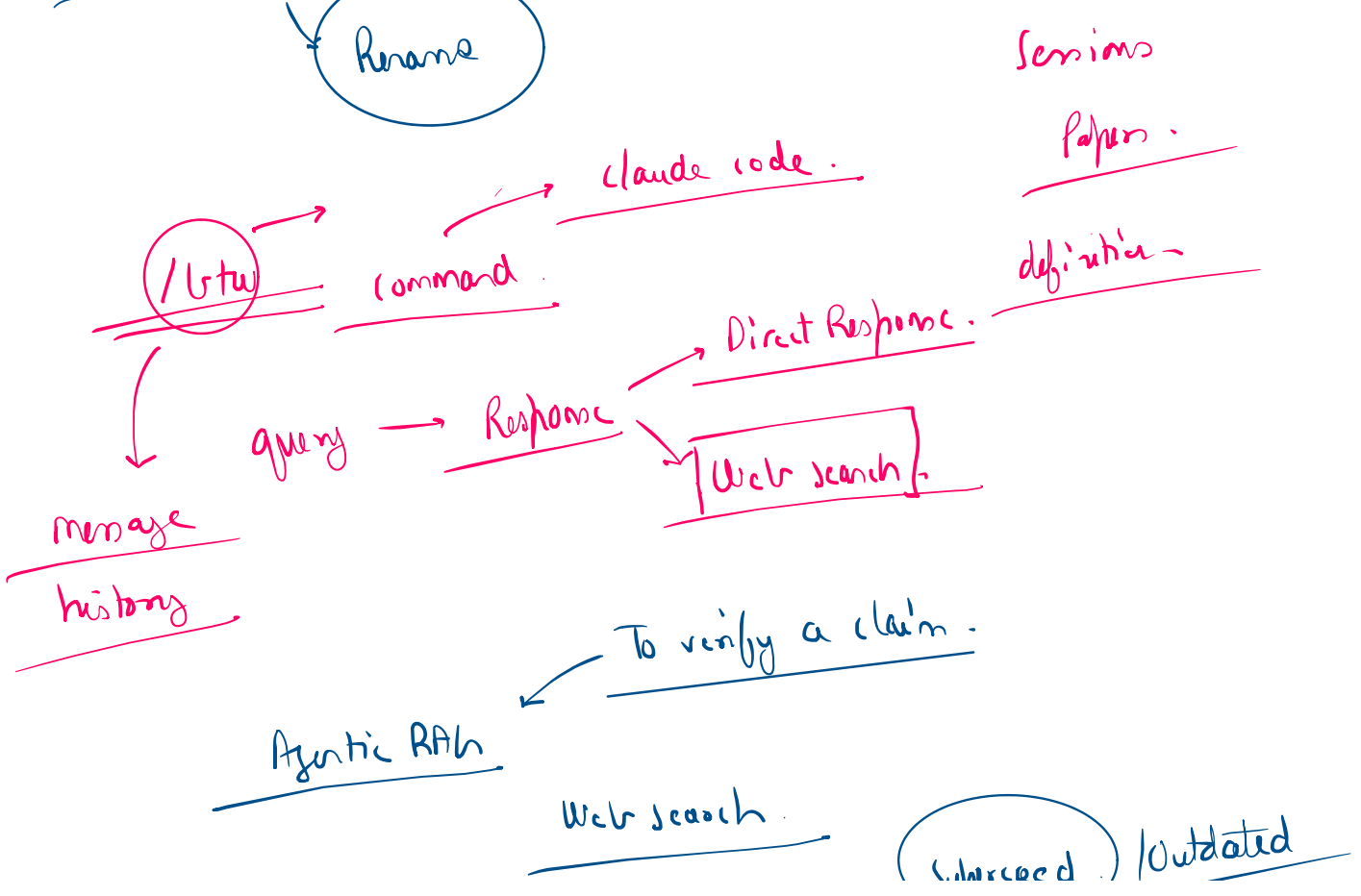
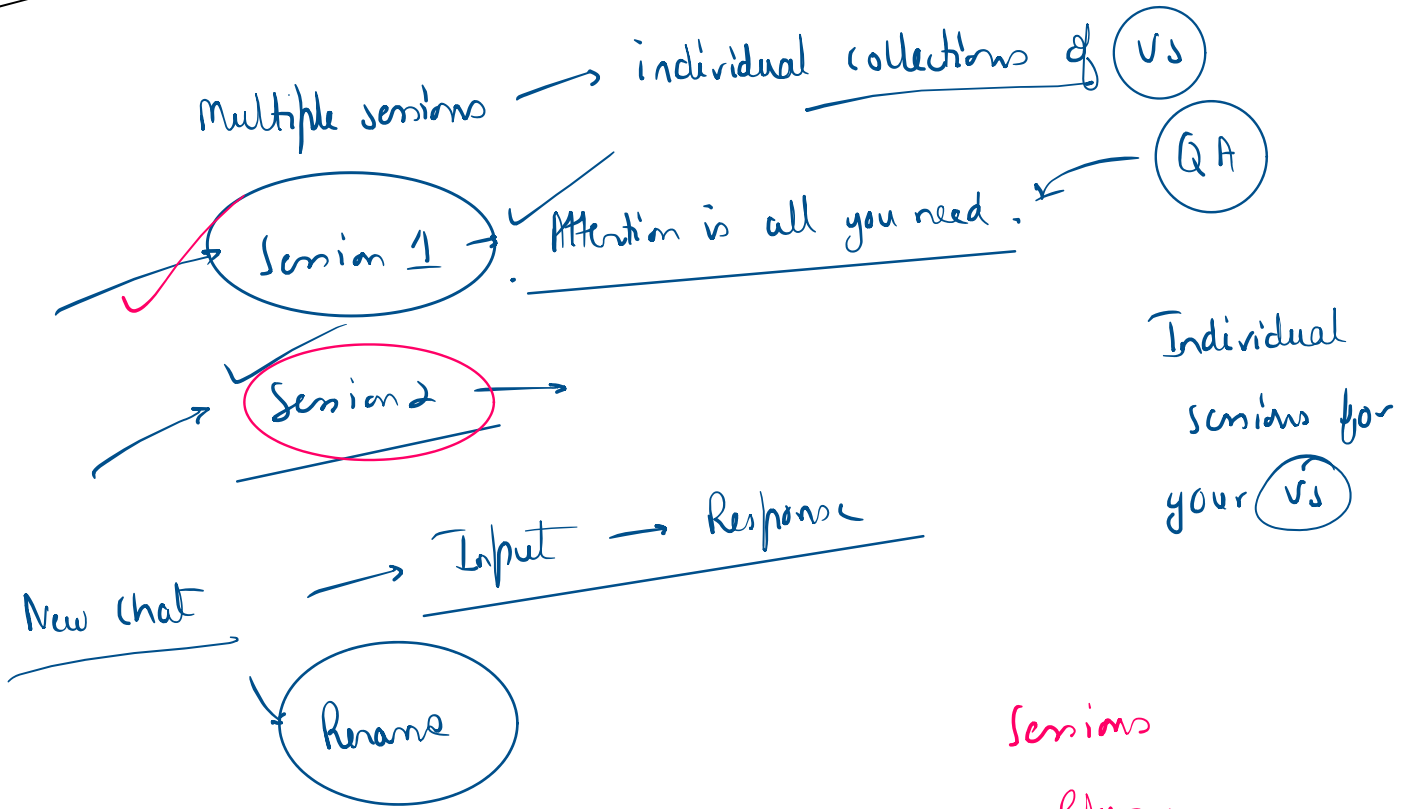
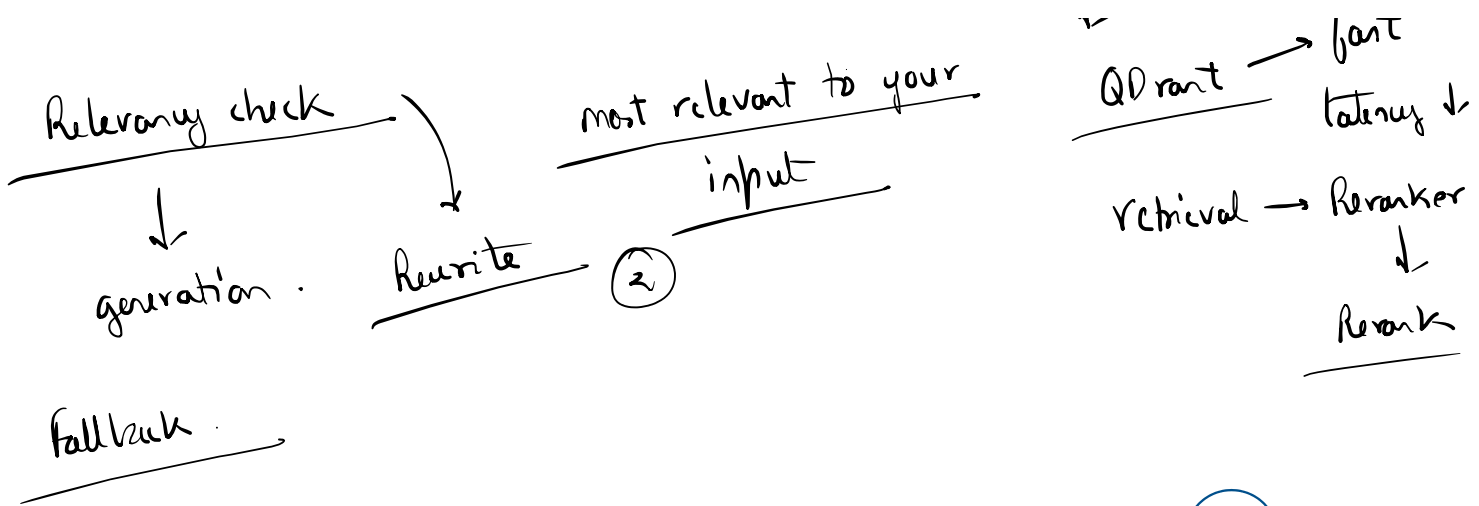


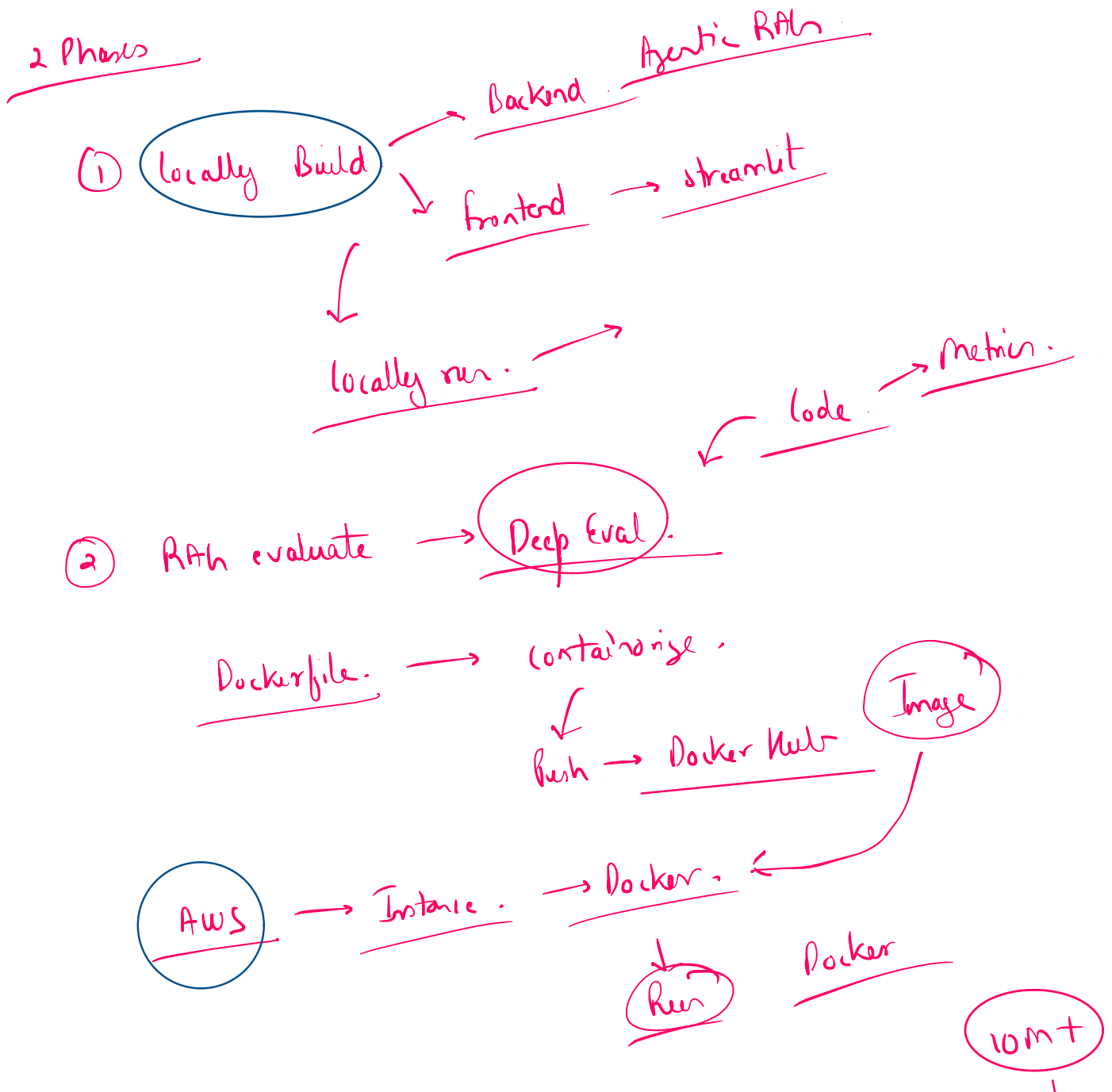
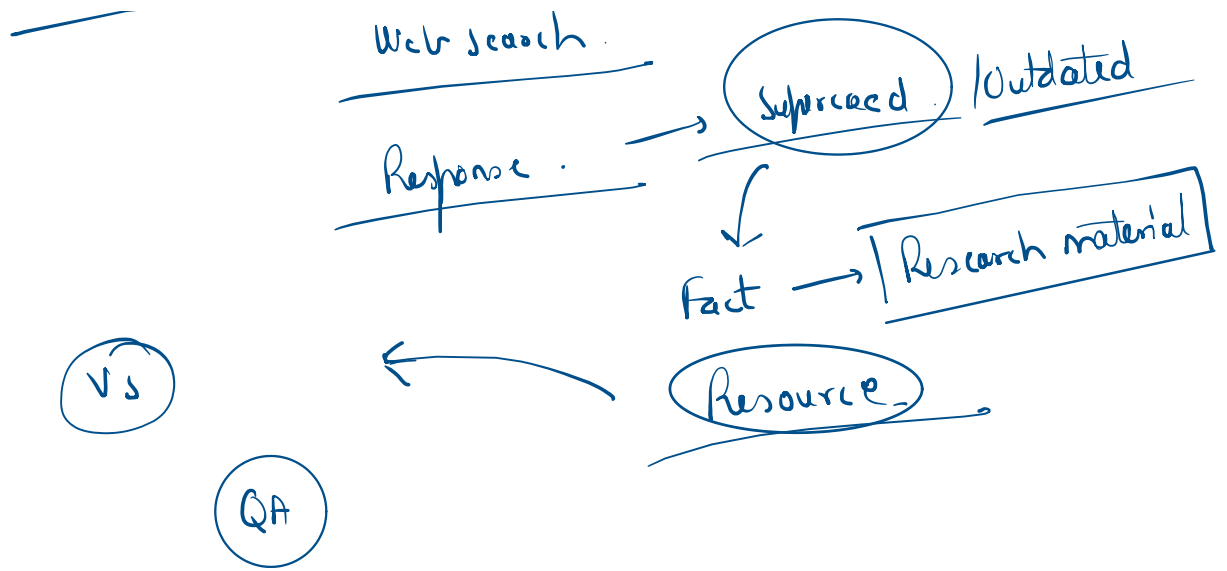
② Remember History and can hold multiple sessions. (Memory)

Stateless application

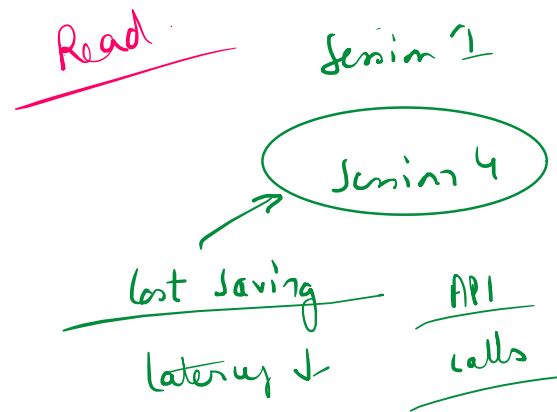
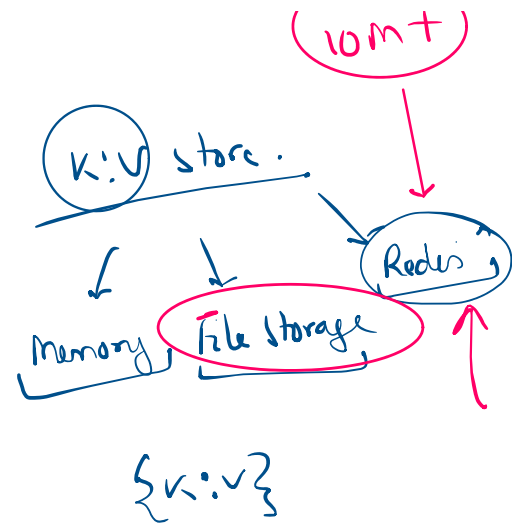
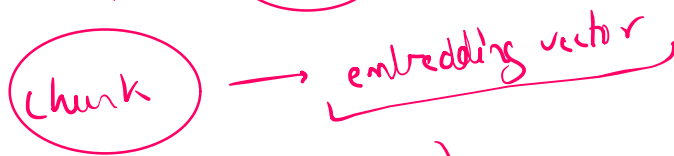
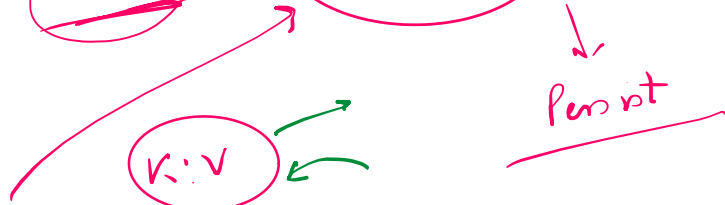
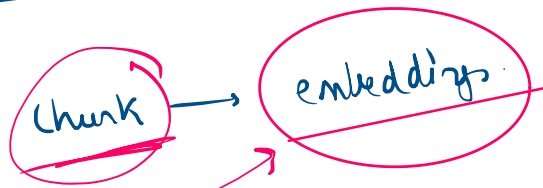
Short term memory







Cached Embeddings: → lacking



Cost Saving
latency ↓

API
calls

Phase 1 → Demo (local system) , code.

Phase 2 → ① RAG Pipeline → Evaluate

② Application → Deployment → AWS

RAGAS (RAG Assessment)

Deep Eval (Evaluate your RAG systems)

LM as a Judge

① Context Precision.

② Context Recall

③ Faithfulness

④ Context Relevancy

⑤ Response Relevancy → RAG Pipeline.

QA Pairs Curate

input-query → (— ,
expected-output → —)

Deep Eval.

Synthesizer (LLM)

Retriever

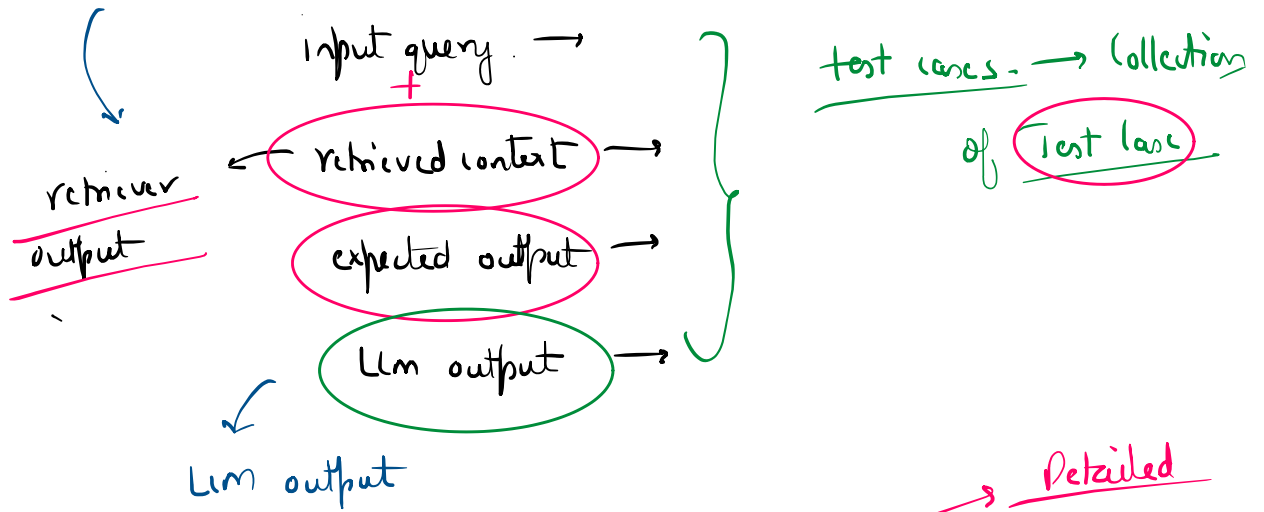
Generator LLM

Deep Eval.

① Golden → QA Pairs. (input query , expected output) ✓

① Golden → QA Pairs. (input query, expected output)

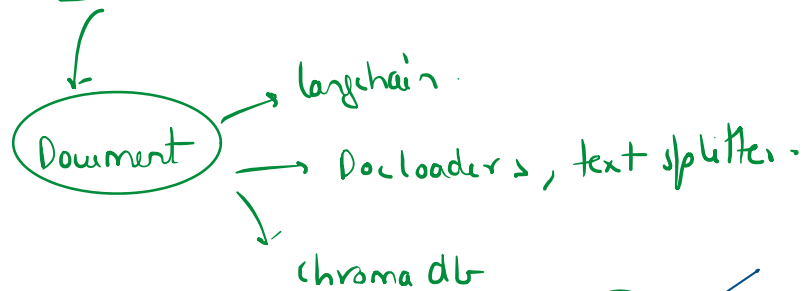
② Test Case → how of Evaluation Dataset → RAG Pipeline



Evaluate API → test cases, metrics → Detailed Report

LLM as a judge → latest models gpt → 5-4 5-4 mini

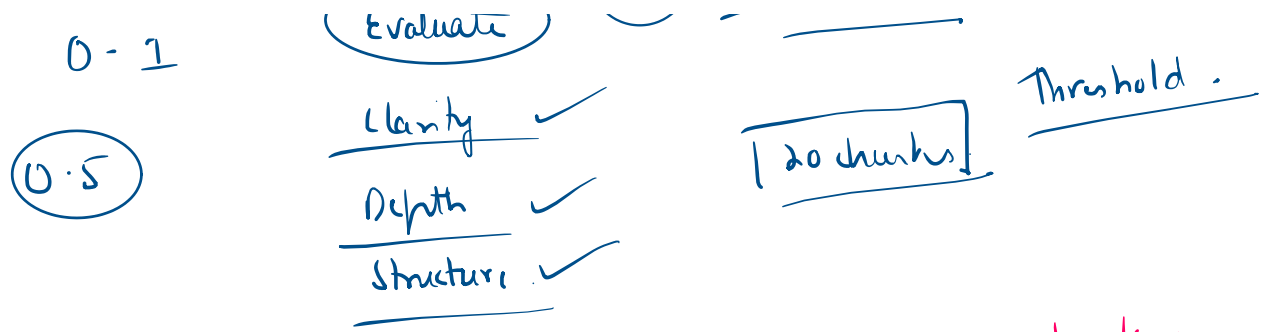
Deep Eval → Synthesizer → LLM



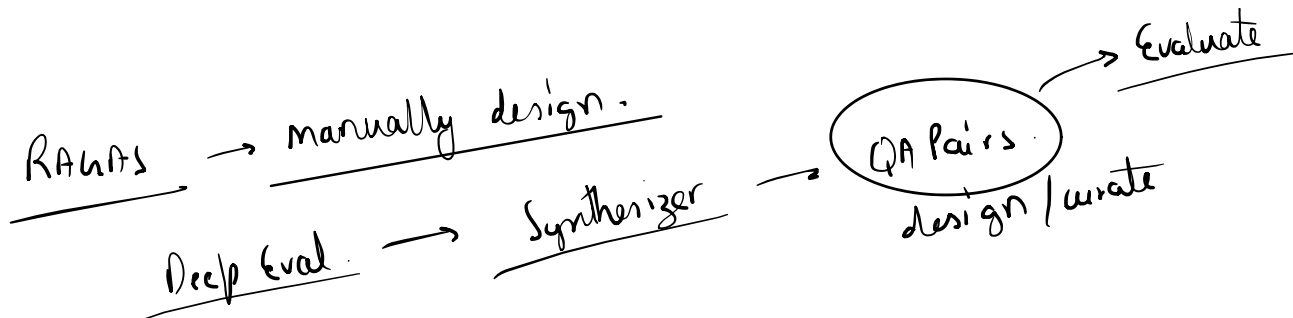
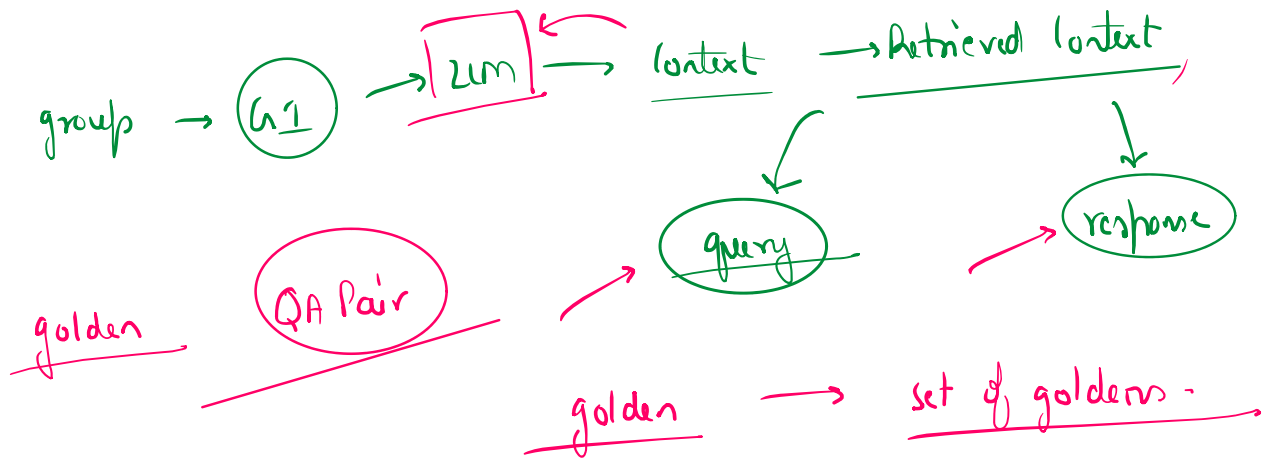
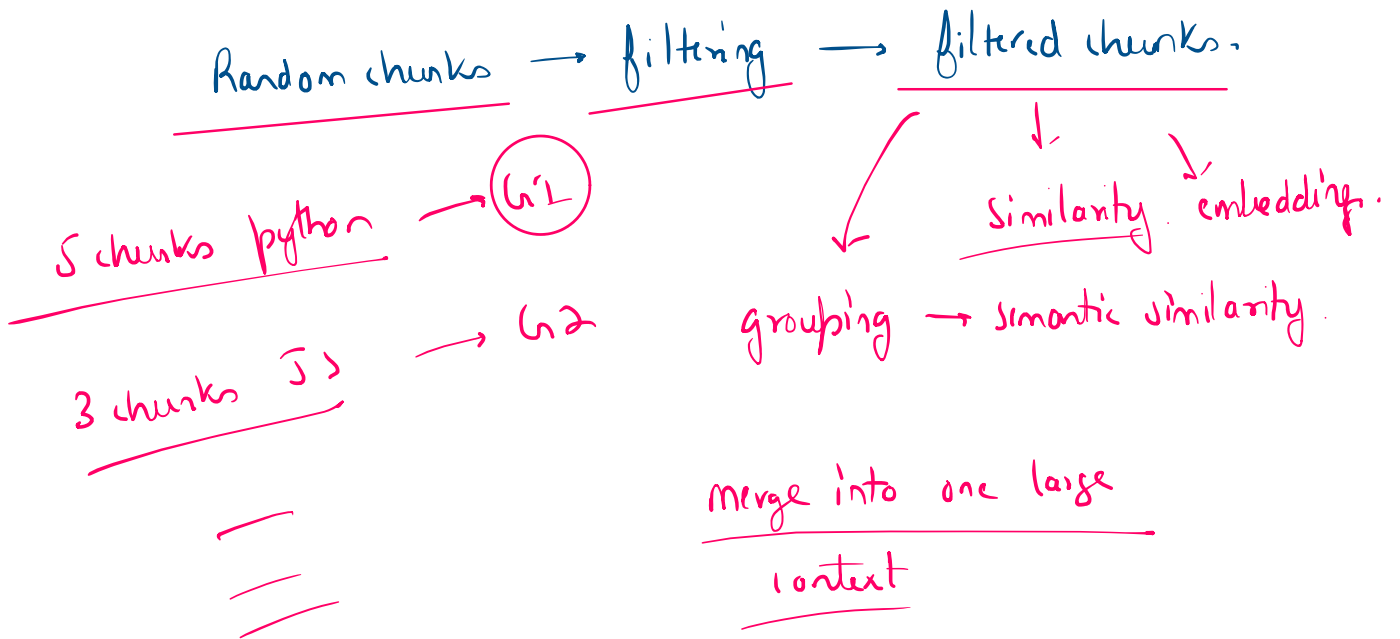
Load, chunking, embedding model → VS ✓

Randomly chunks → Iterations

0-1 Evaluate (LLM) → Score → 50 chunks
Threshold



20 chunks



Deployment

